# Coarse-To-Fine Multiview Anomaly Coupling Network for Hyperspectral Anomaly Detection

Dan Ma, Minghao Chen, Yang Yang, *Member, IEEE*, Beichen Li, Menglong Li, and Yuan Gao

***Abstract*— The fundamental goal of hyperspectral anomaly detection (HAD) is the identification of pixels manifesting substantial deviations in spectral attributes when compared with their neighboring pixels. Nevertheless, the intrinsic attributes of hyperspectral images (HSIs), characterized by their high-dimensional essence and the interdependencies among spectral bands, frequently exert an influence on the efficacy of anomaly detection (AD). Furthermore, current detection algorithms often fall short in harnessing the inherent information encapsulated within HSI, thereby constraining the network's expressive potential. In response to these challenges, we introduce a multiview model tailored that amalgamates both global and local features for HAD. Specifically, the proposed method uses an unsupervised learning-based multiview network to simultaneously conduct feature analysis on both global and local attributes within HSI. The model incorporates a dual-component structure, featuring a global module using axial attention for comprehensive global attribute analysis, and a local module using a convolutional neural network (CNN) with residual connections to capture fine-grained local features. Subsequently, the global–local multiview anomaly coupling mechanism is applied to consolidate the strengths of distinct perspectives, resulting in the ultimate AD outcomes. Experimental evaluations are performed on seven public HSI datasets, demonstrating superior performance of the proposed method in comparison to other state-of-the-art approaches.**

***Index Terms*— Axial attention, convolutional neural network (CNN), hyperspectral anomaly detection (HAD), remote sensing, unsupervised learning.**

## I. Introduction

HYPERSPECTRAL imaging technology represents a crucial instrument for data analysis with extensive applications across the domains of Earth sciences, remote sensing, agriculture, and environmental monitoring [1], [2], [3], [4]. By acquiring spectral reflectance data across a vast array of spectral bands, hyperspectral images (HSIs) adeptly discern and assess distinctions between anomalies and background elements, facilitating the revelation of significant hidden anomalous patterns within the dataset. In stark contrast to the traditional image processing and classification methods that frequently rely on a restricted selection of spectral bands, HSIs draw upon data originating from dozens or even hundreds of spectral bands, thereby affording the capability to capture a more comprehensive array of spectral characteristics. This attribute confers upon HSIs a unique set of advantages, particularly in domains such as anomaly detection (AD) and target recognition, cross-city semantic segmentation, and land cover classification [5], [6].

Hyperspectral AD (HAD) can help people discover unusual phenomena, and then deeply investigate the causes of anomalies and take corresponding measures to cope with them. However, HAD also faces some challenges due to the high dimensionality of HSI, complex spectral features, and correlation between data. Researchers have proposed many innovative methods and techniques to address these challenges and improve the performance of AD. The existing HAD methods can be divided into two categories: traditional methods and deep-learning-based methods.

HAD plays a pivotal role in unveiling uncommon occurrences, subsequently enabling comprehensive investigations into the origins of these anomalies and the formulation of corresponding strategies for mitigation. Nonetheless, HAD encounters formidable hurdles attributed to the high dimensionality of HSIs' data, intricate spectral attributes, and inter-data correlations. To address these challenges and enhance the efficacy of AD, researchers have introduced numerous innovative methodologies and techniques. The extant landscape of HAD approaches can be categorized into two distinct classes: traditional methods and those grounded in deep learning principles.

Reed and Yu [7] initiated research on algorithms for HAD and introduced the renowned Reed–Xiaoli (RX) algorithm. This method postulates that background pixels within a high-dimensional space conform to a multivariate Gaussian distribution, while anomalous pixels deviate from this distribution. It estimates background characteristics using covariance matrices, calculates Mahalanobis distances between individual pixels and the background model, and uses this metric to quantify the spectral distinctions between each pixel and its adjacent background counterparts. Consequently, it accomplishes the task of AD without necessitating prior knowledge of specific target characteristics. Nonetheless, in real-world HSIs, the background information is intricate and does not consistently align with the multivariate Gaussian distribution assumption made by RX. The utilization of the overall image

Manuscript received 20 October 2023; revised 25 January 2024; accepted 31 January 2024. Date of publication 5 February 2024; date of current version 13 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62171318 and Grant 61731003 and in part by the Key Research and Development and Transformation Plan of Science and Technology Program for Tibet Autonomous Region under Grant XZ201901-GB-16. *(Dan Ma and Minghao Chen contributed equally to this work.) (Corresponding author: Beichen Li.)*

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Digital Object Identifier 10.1109/TGRS.2024.3362055

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mean and covariance matrix for background estimation is imprecise. Subsequent researchers have introduced various enhanced algorithms building upon the foundation of RX [8], [9], [10], [11], [12]. In such a case, Molero et al. [8] presented the local RX (LRX) algorithm, which departs from global calculations by engaging in localized computations. This methodology confines its calculations to a small neighborhood of pixels surrounding the test pixel, with each pixel in the image having its unique correlation matrix. Nevertheless, these algorithms, which capitalize on low-order characteristics of HSI for AD, do not account for the nonlinear relationships existing among spectral vectors in HSI. Kwon and Nasrabadi [9] proposed the kernel RX (KRX) as a nonlinear iteration of the RX algorithm. This strategy involves a nonlinear mapping function to project the original data into a high-dimensional feature space, thereby achieving the demarcation of background and anomaly targets within this high-dimensional feature space. Tao et al. [10] introduced an HAD method grounded in fractional Fourier entropy (FrFE), representing a pioneering utilization of frequency-domain Fourier transform (FrFT) for signal representation extraction within an intermediate domain encompassing reflectance spectra and their Fourier-domain information. However, real-world hyperspectral imagery scenarios often present challenges in effectively estimating distributions based on rigid assumptions. To circumvent the necessity of distribution assumptions, numerous scholars have ventured into representation learning [13], [14]. Yuan et al. [15], inspired by sparse representation theory, unveiled the local sparsity divergence-based anomaly detector (LSDAD). This method devises a dual-window model and evaluates the anomaly status of a test pixel by computing its sparse coefficients. Wang et al. [16] used a Potts-based image segmentation algorithm to enhance background uniformity and expanded data matrices with background endmembers, thereby bolstering the effectiveness of low-rank sparse matrix decomposition. Nonetheless, with the growing complexity of scenarios and applications, HSI backgrounds have grown intricate, thereby constraining the detection performance of traditional methods.

In recent years, developments in the field of deep learning have also injected new vitality into the field of HAD [17], [18], [19]. Among these developments, Li et al. [20] have synergized the low-rank representation model with deep learning techniques to develop an innovative HAD network. This network demonstrates proficiency in autonomously learning thresholds and weight parameters through an end-to-end approach, effectively obviating the necessity for manual parameter adjustment. It adeptly exploits the formidable separation capability of the low-rank models combined with the parameter optimization efficiency inherent in deep neural networks. In another notable advancement, Jiang et al. [21] first applied GAN network to HAD and proposed a new unsupervised learning-based method HADGAN for HAD. This method reconstructs conventional autoencoder (AE) networks by introducing a latent discriminator, a consistency enhanced representation constraint, a shrink constraint, and an image discriminator. Fan et al. [22] proposed an unsupervised learning-based robust graph AE (RGAE) detector with the $\mathcal{L}_{2,1}$ norms, a framework that is robust to noise and anomalies during training. Regarding these two unsupervised learning-based approaches, the former method focuses on the reconstruction of the entire image and identifies anomalies through the residual differences between the reconstructed image and the original image. However, this approach overlooks the extraction of local features. In contrast, the latter method delves into the acquisition of local image features through superpixel segmentation but does not address the extraction of global features. Zhong et al. [23] introduced a novel approach for characterizing background–anomaly separability using a generative adversarial network (BASGAN) based on semi-supervised learning for HAD. The primary innovation of this method lies in its explicit constraint on the separability of background and anomalies, achieved by characterizing background spectral samples while avoiding anomalous reconstruction. In a similar vein, Jiang et al. [24] presented a semi-supervised HAD framework built upon a generative adversarial network, aiming to estimate the background distribution in a semi-supervised manner and detect HSI anomalies through the exploration of spectral and spatial features. Meanwhile, Li et al. [25] put forward a semi-supervised learning approach guided by spectral differences and using a graph attention autoencoder (GATE) network for HAD. This approach harnesses GATE to model the inter-sample relationships and introduces spectral sharpening constraints to guide the learning of attention coefficients. It is noteworthy that HAD represents an unsupervised learning task devoid of direct label information for utilization. When applying semi-supervised learning methods to HAD, the presence of errors or noise in the label data can render the semi-supervised learning model sensitive to these inaccuracies, potentially leading to the learning of incorrect patterns and consequently a decline in AD performance.

Detailed scrutiny elucidates that leveraging hyperspectral data for HAD is persistently confronted by a gamut of intricate challenges, primarily stemming from the data's inherent high dimensionality and the complex interrelationships among its spectral components. These impediments not only impinge on the analytical efficiency of hyperspectral data but also constrain its broader applicability in diverse application domains. Pertinently, extant research delineates critical gaps in three principal dimensions.

1) *Enhanced Integration of Spectral and Spatial Attributes:* Hyperspectral imaging, characterized by its elevated spectral sampling rate, facilitates the discernment of subtler spectral nuances and a wealth of spatial data. However, the current landscape of HAD algorithms is narrowly concentrated on either spectral attributes or localized spatial features. This constrained focus may detrimentally affect the algorithms' efficiency and circumscribe their functional applicability in a broad spectrum of application scenarios.
2) *Challenges in Performance of Label-Dependent HAD Models:* Within the domain of label-dependent HAD, the quality and quantity of initial labeling are pivotal in determining the efficacy of the model. Notably, in contexts characterized by intricate data distributions or

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indistinct categorizations, the precision of initial labels becomes crucial, as their inadequacy may culminate in unreliable prediction results. This situation highlights the critical need for augmenting the caliber of labeling and forging algorithms endowed with greater resilience to label scarcity, consequently enhancing overall model performance.

3) *Influence of Spectral Variability on AD Efficacy:* In practical environments, the inevitable spectral variability present in hyperspectral data augments the difficulty of distinguishing between normal and anomalous spectral features, thereby rendering HAD more complex and challenging. Prevalent algorithms in use today exhibit a notable deficiency in fully addressing this spectral variability, consequently constraining their operational efficacy in nuanced, real-world scenarios.

In response to the challenges highlighted above, this study introduces the coarse-to-fine multiview anomaly coupling network for the HAD (MACHAD) algorithm. In contrast to established algorithms, the principal contributions of the proposed method can be summarized as follows.

1) We present an innovative multiview HAD model, founded on the principles of unsupervised learning. This model integrates a sophisticated hybrid network architecture, named Axial-Conformer, designed to simultaneously process and analyze both global and local features inherent in HSI. Within this architecture, the global spatial–spectral feature perception (GSSFP) module capitalizes on axial attention mechanisms, targeting various data dimensions, thus significantly enriching the detection and delineation of intricate relationships and attributes in diverse orientations. In parallel, the local spatial–spectral feature perception (LSSFP) module uses a convolutional neural network (CNN) framework with residual connections, markedly bolstering the model's proficiency in assimilating and interpreting local spatial–spectral features.
2) We propose the multiview anomaly feature coupling (MAFC) module, harnessing a sophisticated nonlinear constrained mechanism for feature coupling. This module adeptly facilitates the synergistic fusion of both local and global features, thereby significantly enhancing the representational prowess of feature.
3) In a pioneering effort, we integrate the axial attention mechanism into the domain of HAD, marking a significant innovation. This mechanism markedly bolsters the model's proficiency in detecting and concentrating on anomalous targets within imagery, exhibiting exceptional adaptability and robustness in the face of intricate and fluctuating spectral data scenarios. This integration adeptly confronts and mitigates the multifaceted challenges inherent in spectral variability.

## II. Related Works

Transformer, introduced by Vaswani et al. [26], marked a significant paradigm shift in the realm of sequence-based tasks, dethroning the traditional dominance of CNN and recurrent neural network (RNN). This innovation led to groundbreaking achievements in various natural language processing domains, including but not limited to machine translation, text generation, and sentiment analysis [27], [28]. Building upon the transformative power of the transformer model, the vision transformer (ViT) [29] emerged as a noteworthy development in the field of computer vision. It adeptly applied the transformer model to tackle image processing challenges, extending its utility and capabilities. Liu et al. [30] proposed a versatile visual backbone network, achieving optimal performance in classification, detection, and segmentation tasks. Chen et al. [31] introduced the image processing transformer, a remarkable innovation that seamlessly integrated the transformer model into the core aspects of visual processing. By harnessing the advantages of large-scale pretraining, they effectively harnessed the transformer's capabilities to excel in diverse image processing endeavors. These tasks encompassed essential functions such as super-resolution, denoising, and rain removal, where the transformer-based approach showcased its prowess. Moreover, given the transformer's exceptional performance in sequence-related tasks, its utilization has extended into video tasks that hinge upon the integration of spatial and temporal data [32]. Yao et al. [33] have proposed a novel multimodal deep learning framework, which accomplishes the task of land use and land cover classification in the field of remote sensing with minimal modifications to the ViT.

Over the recent years, transformer models have garnered substantial attention within the field of HSI processing [34], [35], [36], [37], [38], [39]. Notably, in HSI classification, Zhou et al. [36] introduced the contrastive ViT, offering a pragmatic training model for achieving both high classification accuracy and robust generalization in HSI. Hong et al. [40] have rethought the issue of HSI classification from a sequential perspective and proposed a new backbone network named SpectralFormer, which focuses on the extraction of spectral information. In hyperspectral target detection, Rao et al. [37] proposed a Siamese transformer network for HSI target detection. In this approach, they harnessed cutting-edge transformer networks as the foundational structure for their Siamese network. This adaptation allowed for the extraction of global features from spectral data that encompassed long-range dependencies, resulting in an effective target detection mechanism. In hyperspectral band selection, Liu et al. [38] put forth the BSFormer, a novel transformer-based reconstruction network designed for unsupervised band selection. Within this innovative system, a transformer-based band attention module was introduced, incorporating distinctive band assessment criteria. These criteria included metrics of representativeness and redundancy, working in tandem to maximize the utilization of band representativeness and inter-band correlations for the band selection process. Furthermore, in HAD, Xiao et al. [39] introduced a novel approach known as the spatial–spectral dual-window mask transformer. This approach leverages a transformer network and introduces the innovative concept of "dual-window masks" to efficiently extract background information while adaptively suppressing the reconstruction of anomaly targets. While the methods mentioned above have

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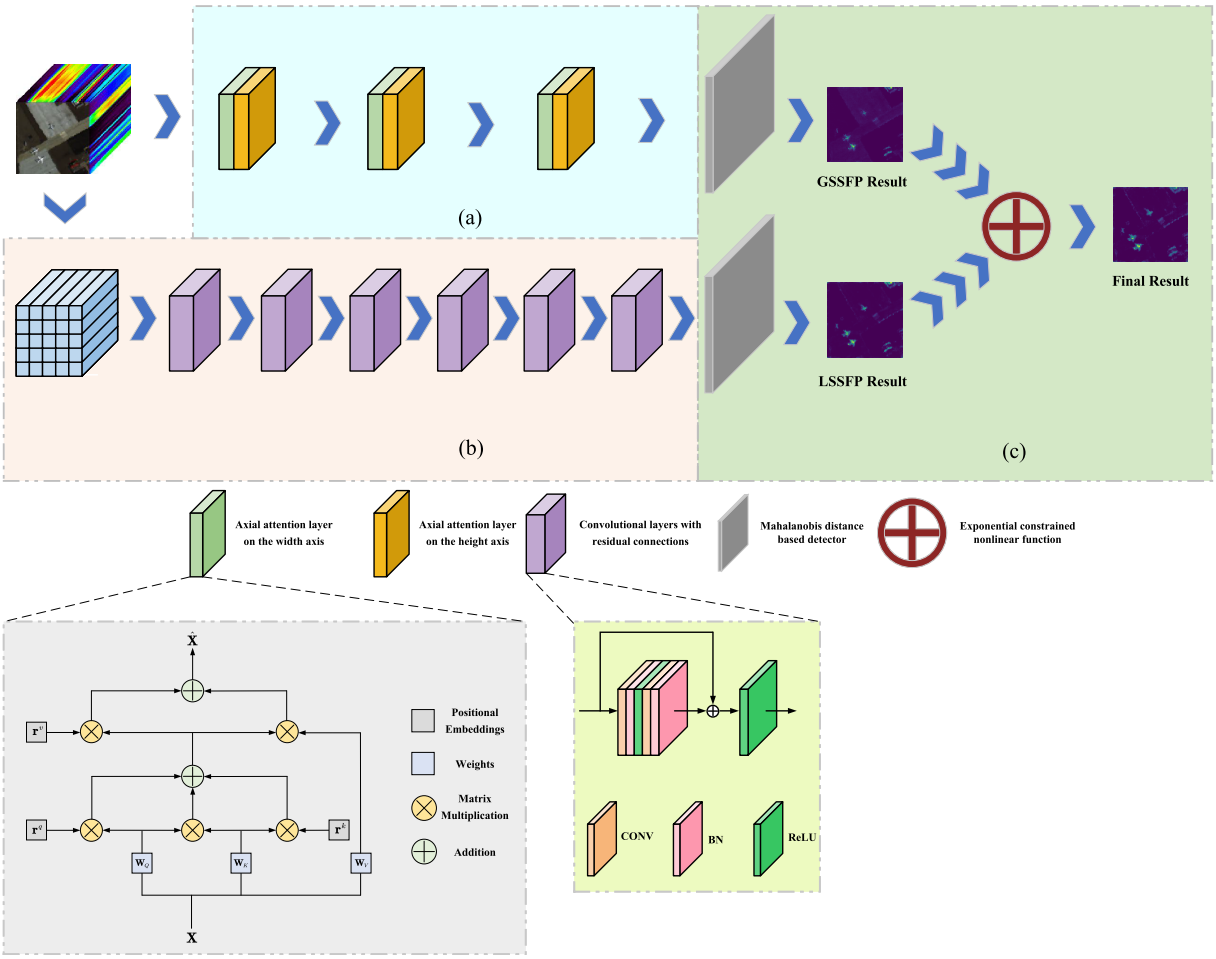



Fig. 1. Schematic of the proposed. (a) GSSFP module. (b) LSSFP module. (c) MAFC module.

demonstrated commendable performance in the domain of HAD, transformer-based approaches often encounter challenges when it comes to providing high-resolution feature information during the decoding process. This limitation, in turn, restricts their ability to accurately predict fine-grained features. With this in mind, to maximize the potential of CNN in the extraction of local features, our research endeavors are focused on a collaborative approach that combines the strengths of both CNN and transformer, thereby gaining an advantage in HAD tasks.

## III. Methodology

In this section, we introduce the proposed MACHAD model and provide an overview of its implementation. The schematic representation of the proposed method is depicted in Fig. 1. This method comprises three core modules: GSSFP, LSSFP, and MAFC. The GSSFP module incorporates an axial attention mechanism to acquire insights into global spatial–spectral features. The LSSFP module is founded on a CNN structure, featuring residual connections, engineered to bolster the acquisition of local spatial–spectral features. The MAFC module is specifically designed to apprehend and establish connections between anomalies found within the global spatial–spectral features obtained from the GSSFP module and those extracted from the local spatial–spectral features through the LSSFP module. This coupling of anomalies is instrumental in fusing complementary information, leading to enhanced detection performance. In the subsequent sections, we will delve into the intricate details of modules GSSFP, LSSFP, and MAFC, along with an exposition of the overall loss function.

### A. GSSFP Module

The traditional self-attention mechanisms often grapple with issues of computational complexity and storage constraints, particularly when processing extensive feature maps. Axial attention, on the other hand, offers an efficient alternative by introducing attention mechanisms along distinct data axes. In the context of image data, axial attention is conventionally applied along the height and width axes. This approach imposes constraints on the attention's focus along each axis, facilitating the capture of long-range dependencies within sequences while simultaneously mitigating the computational overhead. When axial attention is combined with positional encoding techniques and uses relative positional encoding for queries, keys, and values, it enables the model to effectively capture distant positional information.

HSI data are usually high-dimensional and rich in spatial–spectral features, and therefore, an effective attention mechanism is needed to process these data. There may be correlations between different bands, and the use of axial

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attention can help the model learn these correlations and apply them to feature perception and representation, thereby improving the understanding and analysis of HSI.

HSI data typically exhibit high dimensionality and are replete with spatial–spectral features. Consequently, the utilization of an effective attention mechanism becomes imperative in processing such data. It is common for correlations to exist across different spectral bands within HSI data. The integration of axial attention proves advantageous, enabling the model to discern and leverage these correlations for enhanced feature perception and representation. Thus, this approach contributes to an improved understanding and analysis of HSI data.

Let $\boldsymbol{X} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ denote an HSI with $C_{\text{in}}$ channel and $H \times W$ pixels. The calculation output of the traditional self-attention mechanism $\boldsymbol{y} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ is

$$\boldsymbol{y}_{ij} = \sum_{h=1}^{H} \sum_{w=1}^{W} \text{softmax}\left(\boldsymbol{q}_{ij}^T \boldsymbol{k}_{hw}\right) \boldsymbol{v}_{hw} \tag{1}$$

where $\boldsymbol{q} = \boldsymbol{W}_Q \boldsymbol{x}$, $\boldsymbol{k} = \boldsymbol{W}_K \boldsymbol{x}$, and $\boldsymbol{v} = \boldsymbol{W}_V \boldsymbol{x}$ are the projections computed from the input $\boldsymbol{x}$. $\boldsymbol{q}_{ij}, \boldsymbol{k}_{ij}, \boldsymbol{v}_{ij}$ represent the query, key, and value at any position $i \in \{1, \ldots, H\}$, $j \in \{1, \ldots, W\}$, respectively. The projection matrix $\boldsymbol{W}_Q, \boldsymbol{W}_K, \boldsymbol{W}_V \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$ is learnable. The computational complexity of self-attention is proportional to the square of the length of the feature map. Consequently, when confronted with large-scale feature maps, the computational complexity and resource requisites of the self-attention mechanism may become prohibitively impractical.

To address the challenges stemming from computational complexity, a strategic decomposition of self-attention into two axial attention modules is used. One module conducts self-attention operations along the height axis of the feature map, while the other module performs self-attention operations along the width axis of the feature map. This division of the axial attention mechanism effectively emulates the traditional self-attention mechanism, resulting in a substantial enhancement of computational efficiency. The calculation output of the axial attention mechanism on the width axis $\boldsymbol{y} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ is

$$\boldsymbol{y}_{ij} = \sum_{w=1}^{W} \text{softmax}\left(\boldsymbol{q}_{ij}^T \boldsymbol{k}_{iw}\right) \boldsymbol{v}_{iw}. \tag{2}$$

Likewise, the calculation output along the height axis exhibits a similarity.

To harness the relative positional deviations within the image and enhance structural information perception, the proposed method incorporates relative positional encoding into the axial attention mechanism. The calculation output of the axial attention mechanism introducing relative position encoding on the width axis $\boldsymbol{y} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ is

$$\boldsymbol{y}_{ij} = \sum_{w=1}^{W} \text{softmax}\left(\boldsymbol{q}_{ij}^T \boldsymbol{k}_{iw} + \boldsymbol{q}_{ij}^T \boldsymbol{r}_{iw}^q + \boldsymbol{k}_{iw}^T \boldsymbol{r}_{iw}^k\right)\left(\boldsymbol{v}_{iw} + \boldsymbol{r}_{iw}^v\right) \tag{3}$$

where $\boldsymbol{r}^q, \boldsymbol{r}^k, \boldsymbol{r}^v \in \mathbb{R}^{W \times W}$ are the position encoding matrices for the width axis of the feature map. The calculation output on the height axis is similar. By alternately stacking axial attention modules, we can create a computationally efficient model that utilizes axial attention effectively.

In our model, an axial attention block is composed of an axial attention layer on the width axis and an axial attention layer on the height axis. The operation of the axial attention block can be more intuitively represented as

$$\hat{\boldsymbol{X}} = AA_H(AA_W(\boldsymbol{X})) \tag{4}$$

where $AA_H$ and $AA_W$ denote the axial attention layer on the height axis and the axial attention layer on the width axis, respectively. After perceiving global features through several axial attention blocks, the output of the last axial attention block $\hat{\boldsymbol{X}}_{\text{Go}} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ is used as the output of the GSSFP module.

### B. LSSFP Module

The existing HAD methods often adopt a holistic approach in training deep learning models, treating the entire HSI as a unified entity. This approach, however, tends to overlook the learning of essential local features that are pivotal for pixel-level tasks. In addressing this concern, the LSSFP module is specifically designed to incorporate a CNN structure with residual connections. CNNs have demonstrated remarkable performance in various computer vision applications, excelling in the extraction of local features and textural details from images. To facilitate the learning of local features, our approach entails the segmentation of hyperspectral data into patches, allowing the model to effectively perceive and leverage local characteristics inherent in the data.

The LSSFP module comprises a series of stacked residual modules, using a sequence of convolutional layers to systematically process input data and acquire multilevel feature representations. The comparison of input data with the corresponding output data serves as a pivotal mechanism, encouraging the model to discern and learn meaningful feature representations. This orchestrated process ensures the extraction of relevant and discriminative features from the input data.

Specifically, for an input feature map $\boldsymbol{X} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$. It first goes through the patch embedding layer, which is segmented into $N$ patches with the number of spectral bands $C_{\text{in}}$ and the size of $S \times S$ denoted as $\boldsymbol{P} \in \mathbb{R}^{N \times C_{\text{in}} \times S \times S}$, and is used as an input to the LSSFP module. In each convolutional layer within the residual module, a padding operation is implemented to maintain the consistency of the output feature map's dimensions. In addition, a residual connection is established, linking the input and output of the module, which effectively mitigates the issue of gradient vanishing. Subsequently, batch normalization (BN) is applied following the convolutional layer, facilitating the acceleration of network training and acting as a preventive measure against overfitting. To endow the model with the capability for nonlinear mapping learning, the rectified linear unit (ReLU) activation function is used. The previously described operations within the residual module can be more comprehensively visualized as follows:

$$\hat{\boldsymbol{P}} = \text{ReLU}(\text{BN}(\text{Conv}(\text{ReLU}(\text{BN}(\text{Conv}(\boldsymbol{P})))))) + \boldsymbol{P}) \tag{5}$$

where $\hat{\boldsymbol{P}}$ represents the output of the residual module. After perceiving local features through several layers of residual

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module, the output of the last layer of residual module $\hat{\boldsymbol{P}} \in \mathbb{R}^{N\times C_{\text{in}}\times S\times S}$ is re-embedded into the same size as the input image as the output of LSSFP module, denoted as $\hat{\boldsymbol{X}}_{Lo} \in \mathbb{R}^{C_{\text{out}}\times H\times W}$.

### C. MAFC Module

After the reconstruction of the GSSFP module and the LSSFP module, the residual images reconstructed by the two modules can be obtained, respectively. The residual image obtained by the GSSFP module $\Delta\boldsymbol{X}_{\text{Go}}$ is denoted by $\Delta\boldsymbol{X}_{\text{Go}} = |\boldsymbol{X} - \hat{\boldsymbol{X}}_{\text{Go}}| \in \mathbb{R}^{C_{\text{in}}\times H\times W}$. The residual image obtained by the LSSFP module $\Delta\boldsymbol{X}_{Lo}$ is denoted by $\Delta\boldsymbol{X}_{Lo} = |\boldsymbol{X} - \hat{\boldsymbol{X}}_{Lo}| \in \mathbb{R}^{C_{\text{in}}\times H\times W}$.

The initial AD results $\boldsymbol{I}_{\text{Go}}$ and $\boldsymbol{I}_{Lo}$ are obtained using the Mahalanobis-distance-based detector on $\Delta\boldsymbol{X}_{\text{Go}}$ and $\Delta\boldsymbol{X}_{Lo}$, respectively. The formula for the Mahalanobis-distance-based detector is as follows:

$$\boldsymbol{I} = (\Delta\boldsymbol{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Gamma}^{-1}(\Delta\boldsymbol{x}_i - \boldsymbol{\mu}), \quad i = 1, \ldots, H \times W \tag{6}$$

where $\mu \in \mathbb{R}^{C_{\text{in}}\times 1}$, $\Delta\boldsymbol{x}_i \in \mathbb{R}^{C_{\text{in}}\times 1}$ is the $i$th vector of $\Delta\boldsymbol{X}$, and $\Gamma \in \mathbb{R}^{C_{\text{in}}\times C_{\text{in}}}$ are the mean value and covariance matrix of $\Delta\boldsymbol{X}$, respectively.

To facilitate the effective fusion of features obtained from the GSSFP module and LSSFP module, we use a nonlinear operation incorporating an exponential constraint to amalgamate information originating from these disparate feature sources. This approach involves the introduction of an exponential function for assessing the similarity and correlation between various feature sources, while concurrently factoring in the weights and interrelationships among these sources. The integration of an exponential-constrained nonlinear function serves to convert the feature coupling conundrum into a nonlinear optimization problem, with the objective of identifying the most optimal feature matching or fusion outcomes. The final detection result $\boldsymbol{I}_{\text{final}}$ can be computed as follows:

$$\boldsymbol{I}_{\text{final}} = \left(1 - e^{-t\boldsymbol{I}_{Lo}}\right)\boldsymbol{I}_{\text{Go}} \tag{7}$$

where $t$ is a tradeoff parameter. Through the utilization of the nonlinearly constrained feature coupling mechanism, one can effectively control the integration of global and local features within an image. The adjustment of the $t$ value permits a deliberate shift in the weighting between global and local features. Specifically, increasing the $t$ value enhances the prominence of global features, whereas reducing it amplifies the impact of local features in the coupling outcomes. This fine-tuning capability facilitates the harmonization of information from distinct hierarchical levels, whether they pertain to global or local features. As a result, the approach enhances the performance of AD tasks while offering heightened adaptability and resilience.

### D. Overall Objective

The reconstruction loss of the GSSFP module is defined as the $\mathcal{L}_2$ loss between the inputs and outputs of the module as follows:

$$\mathcal{L}_{\text{global}} = \sum_{i=1}^{H\times W} \left\|\boldsymbol{x}_i - \hat{\boldsymbol{x}}_i\right\|_2. \tag{8}$$

**Algorithm 1** Proposed MACHAD Method

**Input:** (1) the original HSI $\boldsymbol{X} \in \mathbb{R}^{C_{in}\times H\times W}$;
(2) the patch size $S \times S$;
(3) the tradeoff parameter $t$;
(4) the number of GSSFP module layers $L_G$;
(5) the number of LSSFP module layers $L_L$.
**Output:** Hyperspectral anomaly detection map.
**Procedure**:
Segment $\boldsymbol{X}$ into patches $\boldsymbol{P} \in \mathbb{R}^{N\times C_{in}\times S\times S}$ via patch embedding layer.
**for** *each epoch* **do**
  **for** *each sample batch* **do**
    Feed $\boldsymbol{X}$ to GSSFP module to get $\hat{\boldsymbol{X}}_{Go} \in \mathbb{R}^{C_{out}\times H\times W}$.
    Feed $\boldsymbol{P}$ to LSSFP module to get $\hat{\boldsymbol{P}} \in \mathbb{R}^{N\times C_{in}\times S\times S}$.
    Backpropagate the total loss (10) to update the learned parameters.
  **end**
**end**
Re-embedding $\hat{\boldsymbol{P}}$ back into $\hat{\boldsymbol{X}}_{Lo} \in \mathbb{R}^{C_{out}\times H\times W}$ via the patch embedding layer.
Get residual image $\Delta\boldsymbol{X}_{Go} = |\boldsymbol{X} - \hat{\boldsymbol{X}}_{Go}| \in \mathbb{R}^{C_{in}\times H\times W}$, $\Delta\boldsymbol{X}_{Lo} = |\boldsymbol{X} - \hat{\boldsymbol{X}}_{Lo}| \in \mathbb{R}^{C_{in}\times H\times W}$.
Get the initial anomaly detection results $\boldsymbol{I}_{Go}$ and $\boldsymbol{I}_{Lo}$ by (6).
Get the final detection result $\boldsymbol{I}_{\text{final}}$ by (7).

The reconstruction loss of the LSSFP module is defined as the $\mathcal{L}_2$ loss between the inputs and outputs of the module as follows:

$$\mathcal{L}_{\text{local}} = \sum_{i=1}^{N}\sum_{j=1}^{S\times S} \left\|\boldsymbol{p}_{ij} - \hat{\boldsymbol{p}}_{ij}\right\|_2. \tag{9}$$

The total loss function of the proposed model is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{global}} + \lambda\mathcal{L}_{\text{local}} \tag{10}$$

where $\lambda$ is the parameter used to control the relative importance of the two losses.

Algorithm 1 summarizes the whole procedure of the proposed MACHAD method.

## IV. Experimental Results and Analysis

### A. Experimental Setup

*1) Datasets:* To evaluate the performance of the proposed method, we conduct validation analyses on seven real hyperspectral datasets: Gulfport, Cat Island, Pavia, Texas Coast-1, Texas Coast-2, Los Angeles, and San Diego. The specific parameters for each dataset are detailed in Table I.

*2) Comparison Methods:* To validate the effectiveness and advancement of the proposed method, it is rigorously benchmarked against nine other representative methods, namely, RX [7], LRX [8], AED [13], CRD [14], FrFE [10], HADGAN [21], BASGAN [23], WGAN [24], and RGAE [22].

The aforementioned baseline methods encompass both the classical approaches and cutting-edge advancements within the

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TABLE I
DATASET DETAILS

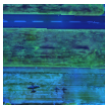
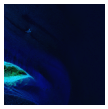
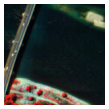
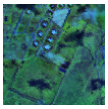
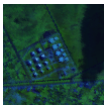
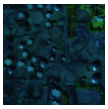
| Dataset Details | Datasets | | | | | | |
|---|---|---|---|---|---|---|---|
| |  |  |  |  |  |  |  |
| | Gulfport | Cat Island | Pavia | Texas Coast-1 | Texas Coast-2 | Los Angeles | San Diego |
| **Sensor** | AVIRIS | AVIRIS | ROSIS | AVIRIS | AVIRIS | AVIRIS | AVIRIS |
| **Resolution** | 3.4m | 17.2m | 1.3m | 17.2m | 17.2m | 7.1m | 3.5m |
| **Spatial Size** | 100×100 | 150×150 | 150×150 | 100×100 | 100×100 | 100×100 | 100×100 |
| **Bands** | 191 | 188 | 102 | 204 | 207 | 205 | 189 |
| **Anomaly Material** | Aircrafts | Ship | Vehicle | Buildings | Buildings | Buildings | Aircrafts |
| **Anomaly Proportion** | 0.60% | 0.08% | 0.28% | 0.67% | 1.55% | 1.70% | 1.34% |

TABLE II
OPTIMAL PARAMETER SETTINGS FOR THE TESTED METHODS

| Methods | Parameters |
|---|---|
| **LRX** | $(w_{in}, w_{out})_{Gulfport} = (11, 13)$<br>$(w_{in}, w_{out})_{Cat\ Island;Pavia} = (11, 21)$<br>$(w_{in}, w_{out})_{Texas\ Coast\text{-}1;Texas\ Coast\text{-}2;Los\ Angeles} = (13, 15)$<br>$(w_{in}, w_{out})_{San\ Diego} = (37, 55)$ |
| **CRD** | $(w_{in}, w_{out}) = (7, 13)$, $\lambda$=$10^{-6}$ |
| **AED** | $\kappa$=25 |
| **FrFE** | $p$=0.8 |
| **HADGAN** | $s$=3, $d$=0.5, $\lambda$=0.5 |
| **BASGAN** | $\lambda$=0.05, $n$=3 |
| **WGAN** | $D$=3, $N$=128, $\lambda$=0.9, $\gamma$=30 |
| **RGAE** | $\lambda$=$10^{-2}$, $S$=150, $d$=100 |
| **Proposed** | $S \times S$= 3 × 3, $t$=5, $L_G$=3, $L_L$=6 |

field of HAD. Specifically, RX, LRX, AED, CRD, and FrFE represent classical methods, while HADGAN, BASGAN, WGAN, and RGAE signify the cutting-edge achievements in HAD. All the methods undergo parameter optimization, and the reported test results represent their peak performance. Table II provides an overview of the optimal parameter configurations for the comparative methods. The RX method operates without the need for specific parameter settings, whereas the LRX method involves two parameters: the inner window size ($w_{in}$) and the outer window size ($w_{out}$). The CRD method encompasses three parameters: the inner window size ($w_{in}$), the outer window size ($w_{out}$), and the regularization parameter ($\lambda$). The AED method uses a single parameter, denoted as the area parameter ($\kappa$). For the FrFE method, the fractional order ($p$) constitutes the sole parameter. HADGAN relies on three parameters: the filter size ($s$), the degree of blurring ($d$), and the weighting parameter ($\lambda$). BASGAN's parameters consist of the separability constraint weight ($\lambda$) and the number of hidden layers ($n$). In contrast, WGAN uses four parameters, including the depth of each subnetwork ($D$), the number of hidden nodes ($N$), and the weighting parameters ($\lambda$ and $\gamma$). RGAE entails three parameters: the weighting parameter ($\lambda$), the number of superpixels ($S$), and the number of hidden layers ($d$). The proposed method incorporates four parameters: patch size ($S \times S$), tradeoff parameter ($t$), the number of GSSFP module layers ($L_G$), and the number of LSSFP module layers ($L_L$). The details of the corresponding parameter selection experiments and the effect of these parameters are conducted and analyzed in Section IV-D.

*3) Evaluation Criterion:* To assess the efficacy and superiority of the proposed approach, this study uses the area under the receiver operating characteristic curve (ROC) metric [41], denoted as $AUC_{(TPR,FPR)}$, to offer a comprehensive evaluation of the detection performance across various HAD methods. This metric's distinctiveness lies in its consideration of the model's detection capability across a spectrum of diverse thresholds, in contrast to the limited focus on the results from a single threshold. It yields a value between 0 and 1, with values closer to 1 signifying superior model performance, while values approaching 0.5 indicate the model's classification ability approximating randomness. Furthermore, this research uses box–whisker plots to assess the discriminative power between background and anomaly in the outcomes of different HAD methods.

*4) Implementation Details:* The proposed method is implemented using Python 3.8 and is based on the PyTorch deep learning framework (version 1.8). The experiments are conducted on a workstation equipped with an Intel Core i7-10700 CPU, a NVIDIA 2080Ti GPU, and 32 GB of RAM. In the proposed framework, the GSSFP module is constructed by stacking three axial attention blocks. The LSSFP module comprises six residual blocks, each operating with a patch size of 9 × 9. Training across all the datasets is conducted over 300 epochs to strike a balance between detection accuracy and computational efficiency. The model optimization uses the Adam optimizer with a learning rate of 0.0001, $\beta_1$ set to 0.9, and $\beta_2$ set to 0.999. The coefficient, used to balance the relative importance of two losses, is established at 1. Furthermore, the tradeoff parameter $t$ for the final detection result is configured to 3.

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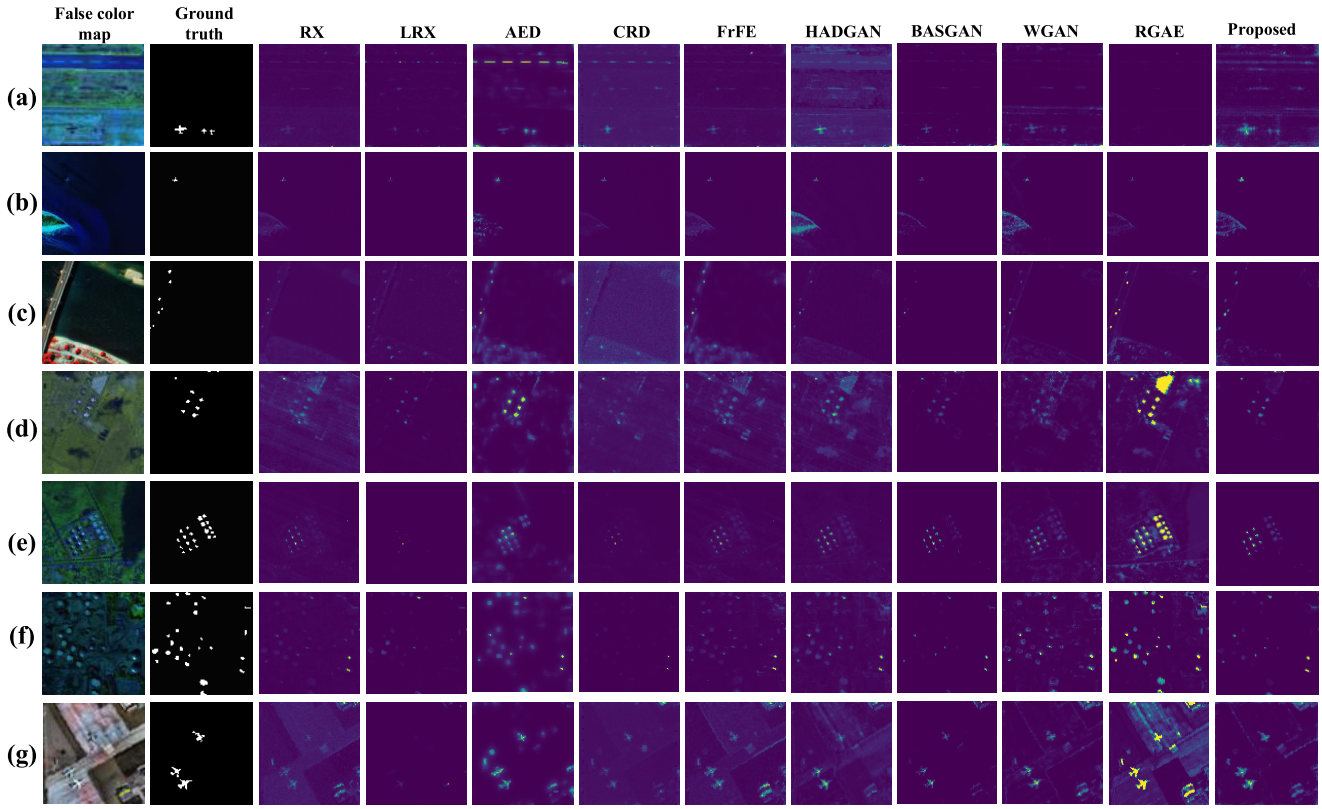



Fig. 2. Color composites, reference map, and detection maps of the compared methods for (a) Gulfport, (b) Cat Island, (c) Pavia, (d) Texas Coast-1, (e) Texas Coast-2, (f) Los Angeles, and (g) San Diego.

### B. Detection Performance

*1) Qualitative Results:* Fig. 2 provides a visual representation of the detection results achieved by various methods across different datasets. The images in the figure clearly demonstrate the disparities in response intensity among the algorithms. Specifically, the RX, LRX, CRD, and BASGAN algorithms exhibit relatively weak responses, resulting in a failure to accurately preserve the target shapes and manifesting varying degrees of underdetection of anomalies. In contrast, the AED, FrFE, HADGAN, and WGAN algorithms produce more pronounced detection outcomes. However, their ability to suppress background information proves less effective, resulting in a noticeable presence of background elements in the detection result images. The RGAE algorithm excels in the satisfactory detection of anomalies on most datasets. Nevertheless, it encounters challenges in efficiently suppressing background interference. Notably, in the case of the Gulfport dataset, the RGAE algorithm experiences visual difficulty in preserving anomalies, virtually losing them in the imagery. Conversely, the proposed method stands out by delivering precise target detection with minimal false positives. This underscores its superiority in both enhancing target detection and effectively suppressing background elements.

The degree of separability between background and anomalies, as determined by all the detection methods, is depicted in Fig. 3. In the figure, blue and red boxes represent the background and anomalies, respectively. The distance separating these two boxes serves as a measure of the extent to which background and anomalies can be distinguished. In addition, the height of the blue boxes is indicative of the effectiveness of background suppression. It is evident from Fig. 4 that the proposed method achieves the highest separability on the Gulfport, Cat Island, and Pavia datasets. Although the separability of the proposed method on the Texas Coast-1, Texas Coast-2, and San Diego datasets is surpassed by the RGAE algorithm, RGAE exhibits greater blue box heights on these datasets. In the case of the Los Angeles dataset, apart from the RGAE and HADGAN algorithms, most algorithms exhibit comparatively lower separability. Nevertheless, the proposed method excels in background suppression when compared with the other approaches.

The ROC curves obtained by various detectors on the seven datasets are presented in Fig. 4. As Fig. 4 illustrates, the ROC curve for the proposed method approaches the upper-left corner on the Gulfport, Cat Island, Pavia, Texas Coast-2, and San Diego datasets. This proximity of the ROC curve to the upper-left corner is indicative of the proposed method achieving superior detection performance on these specific datasets. In addition, the ROC curve of AED closely resembles that of the proposed method on the Texas Coast-1 dataset, implying that both the methods produce similar AUC scores. Moreover, the ROC curve of RGAE on the Los Angeles dataset bears a striking resemblance to that of the proposed method, indicating a comparable potential for attaining similar AUC scores.

*2) Quantitative Results:* To quantitatively assess the performance of the proposed method, Table III presents the AUC values for each method, with the best-performing method

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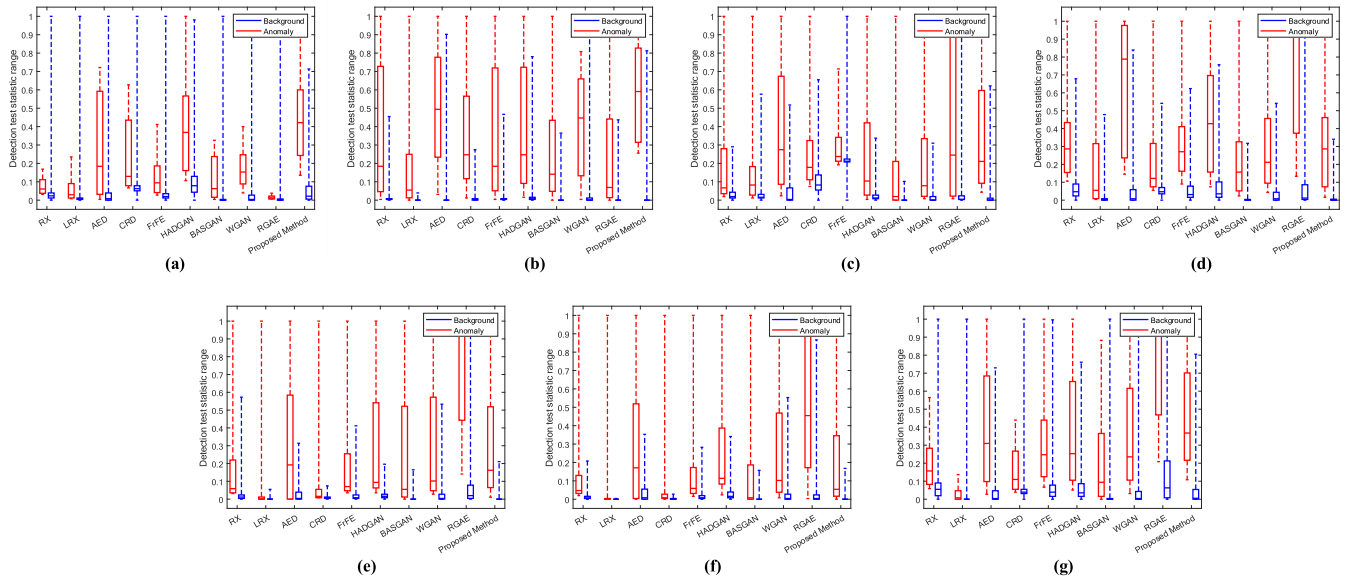



Fig. 3. Separability maps for the ten methods on (a) Gulfport, (b) Cat Island, (c) Pavia, (d) Texas Coast-1, (e) Texas Coast-2, (f) Los Angeles, and (g) San Diego.

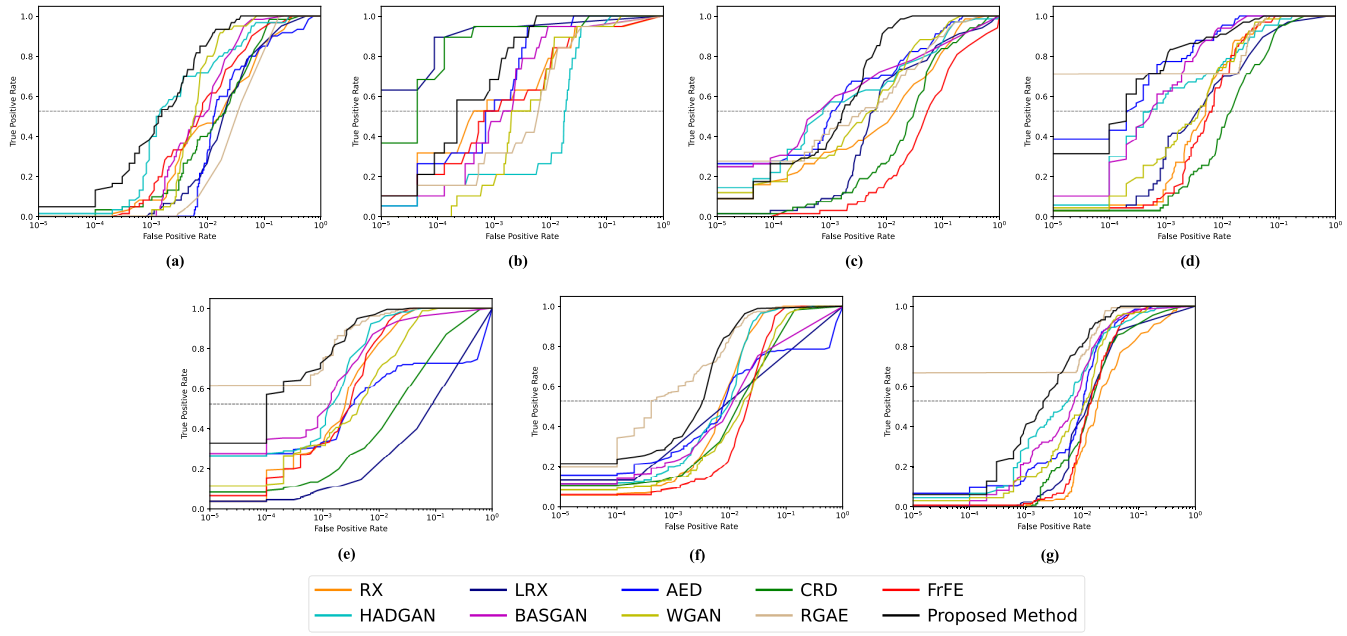



Fig. 4. ROC curves for the ten methods on (a) Gulfport, (b) Cat Island, (c) Pavia, (d) Texas Coast-1, (e) Texas Coast-2, (f) Los Angeles, and (g) San Diego.

highlighted in bold, the second-best results underlined, and the third-best performance displayed in italics. An examination of the table reveals the following.

The proposed method consistently outperforms the majority of comparative algorithms across all the datasets in terms of AUC scores. While the AED algorithm yields a slightly higher AUC score than the proposed method on the Texas Coast-1 dataset, and the RGAE algorithm surpasses the proposed method on the Los Angeles dataset, the proposed method achieves the highest average AUC of 99.59% across all the datasets. This constitutes a substantial improvement of 2.66%, 13.92%, 6.42%, 3.99%, 3.54%, 1.34%, 4.42%, 1.16%, and 1.81% in comparison to other algorithms. Consequently, the proposed method demonstrates heightened robustness and reliability in handling these datasets.

### C. Ablation Study

In this study, we introduce a novel hybrid network architecture denoted as the Axial-Conformer. This architecture leverages a parallel structural framework aimed at optimizing the preservation of both localized features and overarching global representations. By integrating convolutional operations with axial attention mechanisms, it augments the process of feature learning. To comprehensively evaluate the influence of the Axial-Conformer on algorithmic performance, a series of pertinent ablation experiments are thoughtfully devised.

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TABLE III
AUC SCORES OF TPR AND FPR

| Datasets | Methods | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | RX | LRX | AED | CRD | FrFE | HADGAN | BASGAN | WGAN | RGAE | Proposed |
| Gulfport | 0.9536 | 0.9498 | 0.9315 | 0.9650 | 0.9749 | 0.9825 | *0.9854* | 0.9909 | 0.9427 | $\mathbf{0.9925 \pm 0.0009}$ |
| Cat Island | 0.9659 | 0.9736 | *0.9974* | 0.9977 | 0.9666 | 0.9799 | 0.9705 | 0.9866 | 0.9654 | $\mathbf{0.9990 \pm 0.0002}$ |
| Pavia | 0.9520 | 0.9313 | *0.9792* | 0.9259 | 0.8440 | 0.9599 | 0.8849 | 0.9796 | 0.9687 | $\mathbf{0.9967 \pm 0.0004}$ |
| Texas Coast-1 | 0.9906 | 0.9713 | **0.9982** | 0.9691 | 0.9892 | 0.9882 | 0.9975 | 0.9906 | 0.9843 | $0.9972 \pm 0.0006$ |
| Texas Coast-2 | 0.9944 | 0.6795 | 0.7953 | 0.9183 | 0.9953 | *0.9967* | 0.9763 | 0.9886 | 0.9983 | $\mathbf{0.9988 \pm 0.0002}$ |
| Los Angeles | *0.9886* | 0.5661 | 0.8358 | 0.9527 | 0.9749 | 0.9879 | 0.8666 | 0.9699 | **0.9934** | $0.9900 \pm 0.0012$ |
| San Diego | 0.9403 | 0.9250 | *0.9846* | 0.9633 | 0.9786 | 0.9822 | 0.9809 | 0.9838 | 0.9915 | $\mathbf{0.9942 \pm 0.0008}$ |
| **Average** | 0.9693 | 0.8567 | 0.9317 | 0.9560 | 0.9605 | *0.9825* | 0.9517 | 0.9843 | 0.9778 | **0.9959** |

TABLE IV
ABLATION STUDY OF THE AXIAL-CONFORMER USING THE METRIC OF AUC

| Configuration | Datasets | | | | | | | Average |
|---|---|---|---|---|---|---|---|---|
| | Gulfport | Cat Island | Pavia | Texas Coast-1 | Texas Coast-2 | Los Angeles | San Diego | |
| **(a)** | 0.9826 | 0.9972 | 0.9954 | 0.9851 | 0.9930 | 0.9877 | 0.9908 | 0.9903 |
| **(b)** | 0.9875 | 0.9918 | 0.9895 | 0.9942 | 0.9919 | 0.9835 | 0.9886 | 0.9896 |
| **(c)** | 0.9955 | 0.9990 | 0.9967 | 0.9972 | 0.9988 | 0.9900 | 0.9942 | 0.9959 |

Specifically, in Experiment (1), the singular deployment of axial transformer is used for AD, while Experiment (2) focuses on solitary utilization of CNN for the same task. Subsequently, Experiment (3) is conducted to scrutinize AD using the Axial-Conformer. The detailed outcomes of these detection experiments are presented in Table IV for thorough analysis and consideration.

Investigating the performance across seven distinct datasets as delineated in Table IV, when exclusively deploying the axial transformer, the algorithm yielded AUC values of 0.9826, 0.9972, 0.9954, 0.9851, 0.9930, 0.9877, and 0.9908, respectively. Subsequently, upon utilization of CNN, the algorithm demonstrated performance metrics of 0.9875, 0.9918, 0.9895, 0.9942, 0.9919, 0.9835, and 0.9886 for each respective dataset. Notably, the integration of the Axial-Conformer manifested noteworthy enhancements in detection performance, resulting in increments of 1.29%, 0.18%, 0.13%, 1.21%, 0.58%, 0.23%, and 0.34% relative to the axial transformer. In addition, relative to CNN, these enhancements amounted to 0.80%, 0.72%, 0.72%, 0.30%, 0.69%, 0.65%, and 0.56%. Collectively, these findings underscore the efficacy of the proposed hybrid network architecture, in augmenting the performance of AD.

### D. Parameter Analyses

The rigorous examination of parameters assumes paramount importance in the realms of evaluating model robustness, discerning pivotal parameters, and fine-tuning the model's performance. Consequently, this section embarks on a series of parameter analysis experiments on the San Diego dataset, designed to validate the rationality and efficacy of parameter selection. The scope of these parameters encompasses patch size, tradeoff parameters, the number of layers in the GSSFP module, and the number of layers in the LSSFP module.

*1) Evaluations Under Different Patch Sizes:* In light of the method proposed in this article, which partitions the data into input data for the LSSFP module using varying patch sizes facilitated by the patch embedding layers within the LSSFP module, it becomes apparent that the selection of patch size exerts a discernible influence on the outcomes. To meticulously assess the magnitude of this effect and to ascertain the optimal patch size for the method, a comprehensive patch size validation experiment is thoughtfully devised. In the course of the experimental procedure, three distinct patch sizes are configured, specifically $3 \times 3$, $5 \times 5$, and $7 \times 7$, each correlated with unique profiles of AD performance, as elucidated in Fig. 5(a).

Upon meticulous examination of Fig. 5(a), it is manifest that the utilization of a $3 \times 3$ patch size yields the method's apex in detection performance, culminating in an exemplary AUC score of 99.42%. Subsequent in-depth analysis reveals that the subdivision of data into smaller patches plays an instrumental role in empowering the CNN to capture localized features inherent in the input data, thereby resulting in a notable augmentation of the model's proficiency in AD. Given these salient observations, and with an eye on maintaining both procedural fairness and the zenith of experimental performance, the preference for a $3 \times 3$ patch size configuration is consistently upheld throughout the entirety of the experimentation.

*2) Evaluations Under Different Tradeoff Parameter:* To assess the influence of the tradeoff parameter $t$ on algorithmic performance, a dedicated parameter validation experiment is

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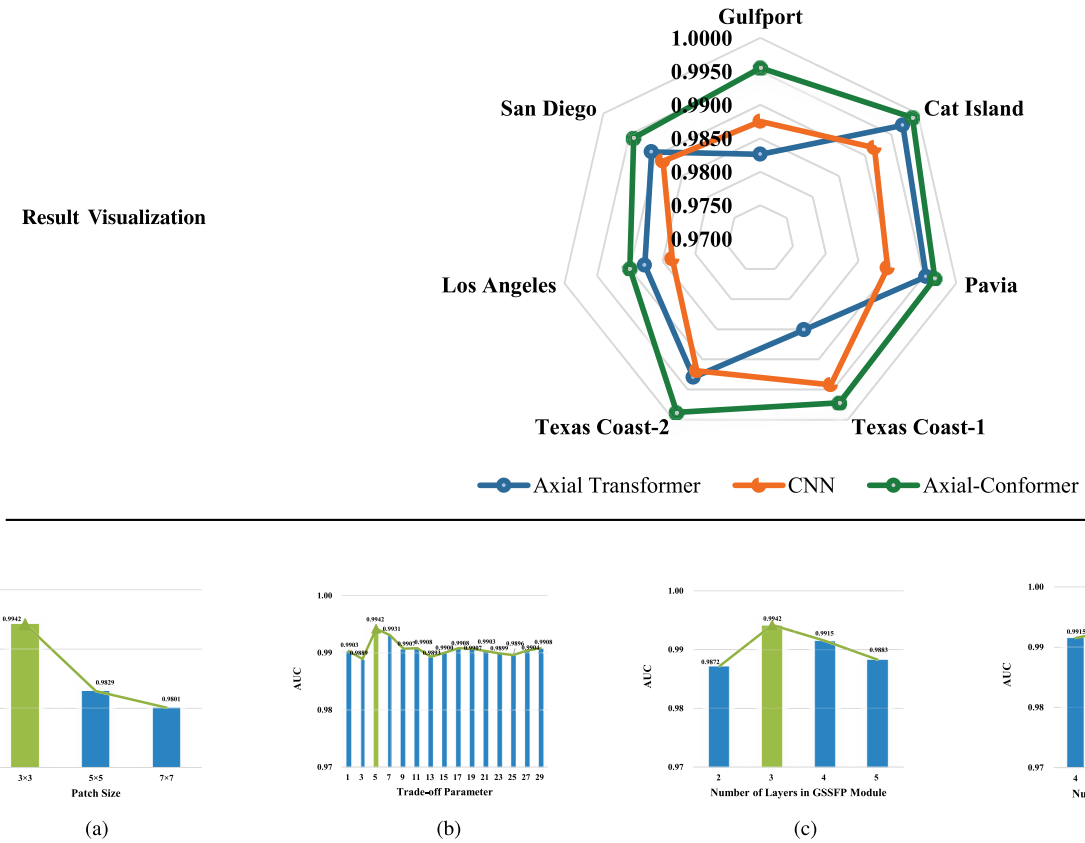



Fig. 5. Analysis of the parameters on the San Diego dataset. (a) Patch size. (b) Tradeoff parameter. (c) Number of layers in GSSFP module. (d) Number of layers in LSSFP module.

meticulously devised for this purpose. Within the framework of the experiment, distinct values of the tradeoff parameter are systematically configured, resulting in a spectrum of performance levels in the context of AD. This deliberate calibration facilitates a comprehensive appraisal of the tradeoff parameter's impact on the algorithm's overall performance, with the resulting empirical observations faithfully rendered in Fig. 5(b).

Within Fig. 5(b), the horizontal axis delineates the values of tradeoff parameter $t$, while the vertical axis conveys the AUC scores, an indispensable metric for the quantitative evaluation of algorithmic efficacy. Remarkably, an apical achievement in detection performance materializes when $t$ attains a value of 5. A deeper scrutiny of the experimental findings unveils that within the construct of the nonlinear constrained feature coupling mechanism, the parameter $t$ exerts a nuanced control over the coupling dynamics between global and local features inherent in the image. As $t$ is augmented, the emphasis on global features is augmented, while conversely, a reduction in $t$ accentuates the role of local features within the coupling framework. Consequently, at $t$ equating to 5, a harmonious amalgamation of information across distinct scales is masterfully achieved, thus culminating in an enhancement of algorithmic performance. In line with the imperative to maintain procedural rigor throughout the entirety of the experiment, the consistent selection of $t$ at 5 is judiciously enforced.

*3) Evaluations Under Different Numbers of GSSFP Module Layers:* To elucidate the influence of the number of layers within the GSSFP module on the algorithm's performance, a rigorous validation experiment is meticulously devised. Within the framework of the experiment, diverse configurations of the GSSFP module layers are systematically established, each corresponding to distinctive levels of AD performance, as graphically depicted in Fig. 5(c).

In Fig. 5(c), the horizontal axis represents the numeric count of layers within the GSSFP module, spanning the range from two to five with an incremental interval of 1. The vertical axis represents the associated AUC performance concerning AD. The graphical representation conspicuously delineates a discernible performance trajectory as the number of layers in the GSSFP module varies from two to five. Notably, the pinnacle of detection performance is discerned when the GSSFP module consists of three layers. An intricate scrutiny of the experimental findings unveils the presence of a nuanced tradeoff between the performance of the GSSFP module and the number of layers. Excessive layering may instigate overfitting, while a paucity of layers may impose limitations on the model's representational capacity. Hence, through comprehensive deliberation and with a steadfast commitment to procedural equity, the selection of three layers for the GSSFP module remained unwavering throughout the course of the experimental endeavors.

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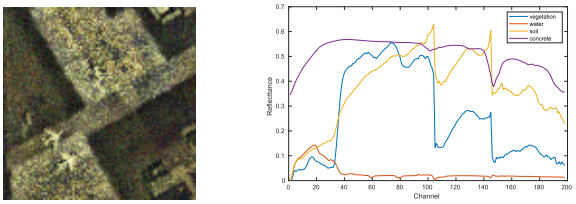
*4) Evaluations Under Different Numbers of LSSFP Module Layers:* To investigate the influence of the number of layers within the LSSFP module on algorithm performance, a meticulous validation experiment is meticulously designed. Within this experimental framework, a variety of diverse layer numbers for the LSSFP module are systematically established, each calibrated to correspondingly manifest distinct levels of performance in AD, with outcomes vividly depicted in Fig. 5(d).

Within Fig. 5(d), the horizontal axis maps the numerical values representing the number of layers within the LSSFP module, spanning from four to eight, while the vertical axis presents the ensuing AUC performance. The graphical exposition manifestly conveys that the optimal level of AD performance is attained when the number of layers within the LSSFP module is judiciously set at six. A comprehensive scrutiny of the experimental results illuminates a noteworthy interplay between the performance of the LSSFP module and the network's depth. Deeper networks, as discerned from the empirical findings, consistently exhibit an enhanced capacity for feature capture and information assimilation. However, it is imperative to acknowledge the delicate tradeoff that deeper networks are potentially susceptible to overfitting. In the spirit of meticulous research and in adherence to the principles of procedural fairness, the configuration of the LSSFP module, consisting of six layers, is scrupulously upheld as a constant across the entirety of the experimentation.

### *E. Model Robustness Analysis*

Due to the inherent spectral variability [42] encountered in hyperspectral data, the task of discerning between normal and abnormal spectral features presents an escalating challenge, thereby intensifying the complexity and demands associated with HAD. To assess the performance of the proposed method in handling spectral diversity, we conduct a model robustness analysis experiment.

The experiment involves simulating spectral variability on the San Diego dataset and comparing it with nine other algorithms. Specifically, the San Diego dataset serves as the foundational source of data for this investigation. To generate mixed pixels, we follow a linear mixing model, merging the spectral attributes of four randomly selected reference endmembers from the United States Geological Survey (USGS) spectral library [43] with the base dataset. In an effort to simulate spectral variability characteristic of real-world scenarios, we introduce random variations in angles and lighting conditions to the endmember spectra. Furthermore, to replicate the presence of noise inherent in spectral measurements, we introduce Gaussian white noise with a signal-to-noise ratio of 25 dB to the mixed pixels. In summary, the proposed method encompasses the consideration of spectral variability across diverse material compositions, environmental influences, and the deliberate introduction of noise. This culminates in the generation of a set of synthetic hyperspectral datasets closely mirroring real-world conditions. These datasets provide an appropriate setting for validating the robustness of the proposed method.




(a) (b)

Fig. 6. False color image of the synthetic data and the spectral curves of the reference endmembers. (a) False color image. (b) Endmembers.

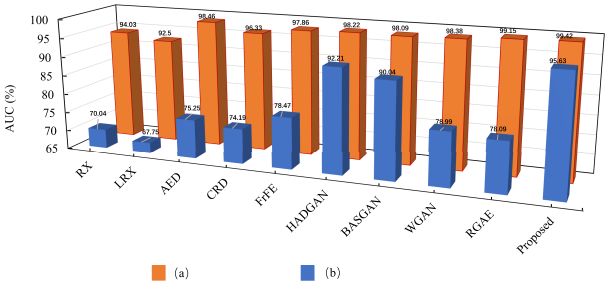



Fig. 7. Quantitative comparison of the detection results on (a) original data and (b) synthetic data by different algorithms.

Fig. 6 displays false color images of the synthetic data and the spectral curves of the reference endmembers.

Fig. 7 presents a quantitative comparison of the AD performance of different algorithms on both the original data and synthetic data. The graphical representation demonstrates that the algorithm proposed in this article attains superior detection performance on both the dataset types. On the synthetic data, it is noteworthy that the proposed method exhibits a marginal decrease in detection accuracy, specifically a reduction of 3.79% in comparison to its performance on the original data. Importantly, this reduction stands in stark contrast to the considerably larger deteriorations observed in the performance of other algorithms, with respective declines of 23.99%, 24.75%, 23.21%, 22.14%, 19.39%, 6.01%, 8.05%, 19.39%, and 21.06%. This result underscores the commendable high-precision detection capability exhibited by our proposed algorithm, even in the presence of spectral variability, thereby underscoring its exceptional robustness in effectively addressing the challenges posed by spectral diversity. Conversely, the detection performance of alternative algorithms exhibits discernible fluctuations. As such, the proposed method demonstrates a significant advantage in terms of model robustness when compared with other algorithms.

In essence, the axial attention mechanism used within the GSSFP module of the proposed algorithm operates autonomously along both the rows and columns of the image, effectively capturing local features. This per-axis processing paradigm significantly contributes to enhancing the model's capacity to discern intricate details within the image's local regions, thereby augmenting its perceptual acuity toward fine-grained particulars. Subsequent to the processing of each axis, the axial attention mechanism adeptly amalgamates the features extracted along diverse axes, thereby comprehensively integrating global contextual information. This strategic approach not only enables the model to preserve its sensitivity

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TABLE V
AVERAGE COMPUTING TIME (IN SECONDS) OF THE COMPARATIVE METHODS

| Datasets | Methods | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | RX | LRX | AED | CRD | FrFE | HADGAN | BASGAN | WGAN | RGAE | Proposed |
| Gulfport | **0.1956** | 42.9308 | 0.2814 | 16.9139 | 14.5476 | 0.2736 | 0.3074 | 0.2994 | 52.8160 | 0.6880 |
| Cat Island | 0.3562 | 90.4966 | 0.4421 | 39.5611 | 30.4066 | **0.2633** | 0.2893 | 0.5233 | 132.2525 | 0.8956 |
| Pavia | **0.2071** | 37.7053 | 0.4073 | 34.9590 | 20.7185 | 0.2545 | 0.2719 | 0.2704 | 104.8476 | 0.6272 |
| Texas Coast-1 | **0.1947** | 47.5262 | 0.2713 | 17.0172 | 14.6841 | 0.2841 | 0.2922 | 0.2475 | 61.8853 | 0.6523 |
| Texas Coast-2 | **0.2065** | 53.0829 | 0.2738 | 17.1392 | 16.8846 | 0.2765 | 0.2663 | 0.2801 | 77.6958 | 0.6453 |
| Los Angeles | **0.1733** | 142.8728 | 0.2606 | 17.2066 | 15.3297 | 0.2955 | 0.2876 | 0.2471 | 77.0959 | 0.6575 |
| San Diego | **0.1708** | 129.2281 | 0.2587 | 17.2016 | 13.4973 | 0.2406 | 0.3656 | 0.2337 | 79.4328 | 0.6178 |
| **Average** | **0.2149** | 77.6918 | 0.3136 | 22.8569 | 18.0098 | 0.2697 | 0.2972 | 0.3002 | 83.7180 | 0.6834 |

to local variations but also fosters a holistic understanding of the overarching structural attributes inherent to the image or dataset. Even when localized spectral variations manifest themselves, the pixel's relationship with the overall image remains unaltered, and the axial attention mechanism maintains focus on this inherent stability. Within the LSSFP module, the CNN exhibits heightened dedication toward capturing the nuanced texture features prevalent within localized image regions. In instances where local pixels undergo spectral variations, they exhibit a noteworthy degree of spectral consistency with neighboring pixels of identical material, while differing from pixels of distinct materials. Exploiting these inherent distinctions with precision, the CNN adeptly extracts spectral features pertaining to anomalous regions. Consequently, in the presence of spectral variations, this algorithm consistently upholds its robustness and successfully accomplishes the task of effective AD.

### *F. Model Efficiency Analysis*

Computational complexity is a key metric for evaluating the efficiency of algorithms. In this section, the computational complexity of different algorithms is analyzed by comparing their runtime on seven hyperspectral datasets, thereby validating the efficiency of the algorithms. The experimental results are presented in Table V. Table V displays the runtime of ten algorithms on seven hyperspectral datasets, with the results recorded in seconds. Table V reveals that the RX algorithm, due to its simple model design, achieves the shortest runtime on six datasets and maintains the shortest average runtime on all the datasets, demonstrating its remarkable execution efficiency. Notably, the last column of Table V draws attention to the considerable decrease in average runtime achieved by the method proposed in this study. It markedly outperforms the LRX, CRD, FrFE, and RGAE algorithms, showing reductions of 77.0084, 22.1735, 17.3264, and 83.0346 s, respectively. In conjunction with the data analysis from Table III, the proposed method surpasses other methods with shorter runtimes, such as RX, AED, HADGAN, BASGAN, and WGAN, in terms of detection accuracy. This indicates that the proposed method ensures high detection accuracy while maintaining a low time cost, sufficiently meeting the timeliness requirements of HAD. Intrinsically, the transformer model network involves a large number of parameters, which increases the algorithm's complexity and runtime. However, the implementation of the axial attention method in this research ambitiously targets a reduction in computational complexity and an enhancement in model efficiency. The axial attention mechanism, in comparison to the traditional attention mechanisms, significantly diminishes computational requirements while concurrently preserving performance levels, thus effectively curtailing runtime overhead. In essence, the method developed in this study successfully attains a sophisticated level of performance, maintaining a judicious balance between computational complexity and advanced functionality.

## V. CONCLUSION

This article introduces a novel coarse-to-fine multiview anomaly coupling model for HAD, achieving an unprecedented comprehensive analysis of hyperspectral data characteristics via the collaborative functioning of GSSFP and LSSFP modules. We pioneer the integration of the axial attention mechanism within the HAD framework, a significant stride in capturing extensive dependencies, complemented by a CNN architecture adept at delineating intricate local details. This cutting-edge coupling strategy synergistically amalgamates the strengths of axial attention and CNN, facilitating the model's adeptness in assimilating complex hyperspectral data attributes on both a global and local scale. In addition, this holistic assimilation of global and local features empowers our model with an enhanced capacity to interpret the nuanced contextual information embedded within the data, thereby markedly elevating the efficacy of AD. The experimental findings highlight the exceptional detection capabilities of the proposed model in HAD tasks, surpassing other state-of-the-art methods. This novel approach contributes a fresh perspective and solution to the field of HAD.

In our future endeavors, we aim to concentrate on the implementation of sophisticated multimodal data fusion techniques in the field of HAD. The continuous evolution of technology equips us with the capability to amalgamate data from a diverse array of sensors, including but not limited to hyperspectral imaging, LiDAR, and synthetic aperture

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radar. This integrative approach does not merely expand the dimensionality of our data but also substantially broadens our analytical horizon, providing a more nuanced and profound understanding of AD. Such enhancement significantly bolsters the precision in pinpointing and localizing anomalies, thereby propelling the advancement of the HAD.

## REFERENCES


[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] T. A. Carrino, A. P. Crósta, C. L. B. Toledo, and A. M. Silva, "Hyperspectral remote sensing applied to mineral exploration in southern peru: A multiple data integration approach in the chapi chiara gold prospect," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 64, pp. 287–300, Feb. 2018.

[3] T. Ad ao et al., "Hyperspectral imaging: A review on UAV-based sensors, data processing and applications for agriculture and forestry," *Remote Sens.*, vol. 9, no. 11, p. 1110, Oct. 2017.

[4] M. Xu et al., "Regionally and locally adaptive models for retrieving chlorophyll-a concentration in inland waters from remotely sensed multispectral and hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4758–4774, Jul. 2019.

[5] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[6] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[7] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[8] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[9] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[10] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[11] L. Ren, L. Zhao, and Y. Wang, "A superpixel-based dual window RX for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 7, pp. 1233–1237, Jul. 2020.

[12] X. Yang, X. Huang, M. Zhu, S. Xu, and Y. Liu, "Ensemble and random RX with multiple features anomaly detector for hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[14] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[15] Z. Yuan, H. Sun, K. Ji, Z. Li, and H. Zou, "Local sparsity divergence for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1697–1701, Oct. 2014.

[16] S. Wang, X. Wang, Y. Zhong, and L. Zhang, "Hyperspectral anomaly detection via locally enhanced low-rank prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 6995–7009, Oct. 2020.

[17] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS$^3$LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[18] X. Shen, H. Liu, J. Nie, and X. Zhou, "Matrix factorization with framelet and saliency priors for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504413.

[19] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. J. Plaza, "PDB-SNet: Pixel-shuffle downsampling blind-spot reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511914.

[20] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.

[21] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[22] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.

[23] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[24] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[25] K. Li et al., "Spectral difference guided graph attention autoencoder for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–17, 2023.

[26] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.

[27] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[28] T. B. Brown et al., "Language models are few-shot learners," 2020, *arXiv:2005.14165*.

[29] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[30] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[31] H. Chen et al., "Pre-trained image processing transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12294–12305.

[32] J. Wang and L. Torresani, "Deformable video transformer," 2022, *arXiv:2203.16795*.

[33] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[34] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8989–9003, Nov. 2023.

[35] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.

[36] H. Zhou, X. Zhang, C. Zhang, and Q. Ma, "Vision transformer with contrastive learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[37] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[38] Y. Liu, X. Li, Z. Xu, and Z. Hua, "BSFormer: Transformer-based reconstruction network for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[39] S. Xiao, T. Zhang, Z. Xu, J. Qu, S. Hou, and W. Dong, "Anomaly detection of hyperspectral images based on transformer with spatial–spectral dual-window mask," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1414–1426, 2023.

[40] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[41] P. A. Flach, J. Hernández-Orallo, and C. Ferri, "A coherent interpretation of AUC as a measure of aggregated classification performance," in *Proc. ICML*, 2011, pp. 657–664.

[42] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[43] R. Kokaly et al., "USGS spectral library version 7 data: US geological survey data release," United States Geological Surv. (USGS), Reston, VA, USA, Tech. Rep., 61, 2017.


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